

A Feasibility-Aware Evaluation Protocol for HUBO/QUBO UAV Obstacle-Avoidance with Visibility Cost

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Abstract

Quantum-assisted UAV path planning has recently been studied using QAOA-style obstacle-avoidance formulations, but a low-energy binary sample is not necessarily a valid UAV trajectory. This paper presents a feasibility-aware evaluation protocol on three fixed 8×8 , $L = 20$ UAV grid-planning instances with obstacle avoidance, a line-of-sight visibility cost, and a higher-order temporal buffer-risk term. The instances share the start, target, horizon, and objective weights while varying obstacle count and placement; L denotes discrete planning layers, not physical seconds. The model uses binary variables $x_{t,v}$ indicating whether the UAV occupies cell v at time layer t and forms a HUBO objective with one-hot, start, motion, obstacle, visibility, terminal-goal, and sustained near-obstacle proximity terms. The primary six-obstacle instance has 1,280 original binary variables and 118,344 polynomial terms, including 4,392 cubic monomials. We compare A*, RRT-best, a QUAU-style QAOA candidate-path selector, native trajectory-level simulated annealing (SA), the classical `neal` BQM sampler after HUBO-to-QUBO reduction, and a CP-SAT baseline that exactly solves an integer-scaled hard-feasible objective. CP-SAT returns OPTIMAL on all three instances. On the primary instance, it certifies that the best observed SA path is globally optimal for the implemented scaled objective, with implementation HUBO energy -20752.43 , soft cost 247.57, path length 14, six turns, and 85% visibility. Across the primary, two-obstacle, and three-obstacle instances, SA full-task success is 16%, 84%, and 66%, whereas `neal` full-task success is 0% in all three. These fixed-instance sensitivity results show why feasibility-aware decoding, exact baselines, and quadratization effects matter; they do not establish quantum speedup or distribution-level performance.

Keywords: HUBO, QUBO, feasibility-aware evaluation, UAV path planning, obstacle avoidance, visibility-aware planning, CP-SAT, QAOA, quantum-compatible optimization.

Introduction

UAV path planning in cluttered environments requires at least two safety properties: the path must avoid obstacles, and the decoded plan must be executable under the allowed motion model. When perception or tracking is relevant, the planner may also prefer cells with line of sight to the target.

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Recent quantum-assisted work such as QUAU (Innan et al. 2025) formulates UAV obstacle-avoidance path planning using QAOA-style optimization and compares against A* and RRT. QUAU is therefore an important closest reference for quantum-assisted UAV navigation. However, a QAOA, annealing, or QUBO solver returns a binary sample, and the sample must be decoded and checked before it can be called a valid path.

This paper uses a deliberately compact grid so that modeling and solver behavior can be inspected directly. Rather than claiming quantum speedup, it defines a reproducible HUBO/QUBO-compatible protocol on one primary instance and two fixed obstacle-layout sensitivity instances, separates energy minimization from path validity, and compares classical, quantum-inspired, and quantum-compatible solver pathways.

Problem statement and scope

The planning problem considered here is a static, discrete visibility-aware path-planning abstraction. A UAV must move from a fixed start cell to a fixed target/goal cell through legal grid transitions while avoiding obstacle cells. Visibility is represented as a soft line-of-sight cost from each candidate UAV cell to the fixed target cell. Thus, this paper does not solve continuous quadrotor control, full moving-target tracking, or camera field-of-view control. The abstraction is intentionally small enough that the encoded binary model, decoded paths, feasibility failures, and exact classical baseline can be inspected in detail. The time horizon L is a design parameter giving the number of discrete planning layers. It is not assumed to be seconds unless a physical sampling time Δt is separately assigned.

The contributions are as follows. First, we give a complete binary decision-variable formulation for UAV grid planning with obstacle avoidance and a visibility cost. Second, we identify the real source of higher-order structure: a temporal buffer-risk term, not the static line-of-sight cost. Third, we introduce a feasibility-aware evaluation protocol that reports structural feasibility, goal arrival, and full-task success separately. Fourth, we implement a same-platform QUAU-style QAOA candidate-path baseline, clearly not as official QUAU code but as a reproducible reference following the published workflow. Fifth, we add a CP-SAT implementation that certifies global optimality for an integer-scaled hard-feasible

objective. Finally, we repeat the full comparison on two additional fixed obstacle maps and report seeds and 95% Wilson intervals for the stochastic success rates.

The novelty is narrowly the feasibility-aware evaluation layer around a quantum-compatible UAV HUBO/QUBO formulation. The same decoded paths are compared under graph-search, candidate-path QAOA, native trajectory-SA, quadratized BQM sampling, and exact CP-SAT, with validity and goal arrival separated from energy. The paper claims neither a faster planner than A* or CP-SAT nor quantum advantage; without exact baselines and decoded-path checks, low-energy samples can be misleading.

Related Work

Classical and visibility-aware UAV planning

Classical UAV planning spans graph search, sampling-based planning, trajectory optimization, and perception-aware control. A* remains a standard shortest-path reference on discretized maps (Hart, Nilsson, and Raphael 1968), while RRT provides a widely used sampling-based alternative (LaValle 1998). For target-tracking and perception-aware flight, Chen *et al.* formulate moving-target tracking in cluttered environments (Chen, Liu, and Shen 2016); Hausman *et al.* study occlusion-aware multi-robot 3D tracking (Hausman *et al.* 2016); Penin *et al.* combine aggressive target tracking with collision and occlusion avoidance (Penin, Robuffo Giordano, and Chaumette 2018); and Falanga *et al.* incorporate perception objectives into model-predictive control (Falanga *et al.* 2018). Wang *et al.* extend visibility-constrained tracking to multiple UAVs (Wang *et al.* 2022), while Masnavi *et al.* derive a real-time multi-convex MPC for occlusion-free target tracking with collision avoidance (Masnavi *et al.* 2022). These continuous-control works motivate the visibility term used here, but our problem is deliberately a static discrete evaluation instance rather than a moving-target control formulation.

Quantum and QUBO formulations for UAV/drone optimization

Quantum optimization for routing has developed rapidly; QUBO provides a standard binary optimization representation (Glover, Kochenberger, and Du 2019), while QAOA is a widely used variational gate-based approach (Farhi, Goldstone, and Gutmann 2014). A systematic review by Osaba *et al.* surveys early quantum-routing formulations and highlights recurring issues of encoding size, constraints, and fair classical comparison (Osaba, Villar-Rodriguez, and Oregi 2022). Several UAV-specific studies are especially relevant. Huang *et al.* use variational quantum algorithms, including QAOA and VQE, to select collision-avoiding routes for multiple UAVs from predefined alternatives (Huang *et al.* 2022). Davies and Kalidindi formulate constrained multi-UAV mission planning as MILP and QUBO and study quantum annealing, QAOA, and VQE routes (Davies and Kalidindi 2024). Hua *et al.* investigate QUBO-based drone route planning with quantum/digital annealers and compare against Gurobi on wind-influenced urban routing instances (Hua *et al.* 2024). Tarquini *et al.* compare QUBO formulations of

the Drone Delivery Packing Problem under quantum annealing, simulated annealing, and classical global optimization, emphasizing both qubit-reduction opportunities and feasibility/scalability limits (Tarquini *et al.* 2026).

More recent UAV-specific frameworks move closer to the present setting. Q4DR combines a QAOA clustering stage with annealing-based drone routing under asymmetric costs, forbidden arcs, and charging constraints (Osaba *et al.* 2025). QUAV uses QAOA-style optimization for single-UAV obstacle-avoidance path planning and reports both simulation and quantum-hardware experiments; it is the closest prior work to our candidate-path QAOA baseline (Innan *et al.* 2025). Qiu *et al.* formulate multi-target UAV path planning as QUBO and solve it using a coherent Ising machine, providing another direct example of binary optimization for UAV navigation (Qiu *et al.* 2025). Barbeau *et al.* formulate 1D and 3D target tracking, including obstacle-distance constraints, as constrained quadratic models for quantum annealing (Barbeau, Janabi-Sharifi, and Masnavi 2025). These studies demonstrate multiple quantum-compatible routes for UAV routing, collision avoidance, and tracking, but they optimize different abstractions and generally do not center evaluation on decoded structural feasibility under a common native objective.

QUBO path finding, multi-agent planning, and the gap addressed here

Beyond UAV-specific work, recent multi-agent path-finding studies are relevant to encoding and scalability. Gerlach *et al.* propose an optimal hybrid quantum-classical MAPF method based on branch-and-cut-and-prize with iteratively solved QUBO conflict subproblems (Gerlach *et al.* 2025). Gonzalez Villasmil proposes a robotics-oriented time-indexed QUBO with BFS-based logical preprocessing, adaptive penalties, and time-window decomposition for multi-robot grid planning (González Villasmil 2026). These works reinforce two lessons that directly motivate our evaluation protocol: path-planning encodings can grow rapidly with space-time discretization, and solver outputs must be interpreted relative to hard feasibility and strong classical references.

Among the works reviewed above, we did not find a study combining a single-UAV time-indexed HUBO with a static line-of-sight cost, a genuinely cubic sustained near-obstacle term, HUBO-to-QUBO conversion, decoded feasibility/goal/full-success reporting, and a certified CP-SAT reference on the corresponding hard-feasible objective. The contribution is a controlled evaluation protocol showing when favorable binary-optimization energies do or do not represent executable UAV paths, not a quantum-advantage claim or broad instance suite.

Discrete Planning Model

Let $G = (V, E)$ be an $N \times N$ grid graph. In the experiments $N = 8$, $|V| = 64$, and time layers are indexed $t = 0, \dots, L-1$ with $L = 20$. Each time layer is one discrete trajectory frame; a larger L gives the planner more room for detours or hovering but increases the number of variables linearly. Let $O \subset V$ be the obstacle set, $s \in V$ the start cell, and $g \in V$

the target cell. The permitted next-cell set is

$$\mathcal{N}(u) = \{u\} \cup \{v \in V : (u, v) \in E\}, \quad (1)$$

where the self-loop represents hovering.

The binary state variable is

$$\begin{aligned} x_{t,v} &\in \{0, 1\}, \\ x_{t,v} = 1 &\Leftrightarrow \text{the UAV is at cell } v \text{ at time } t. \end{aligned} \quad (2)$$

A valid binary solution decodes to a trajectory $\pi = (v_0, \dots, v_{L-1})$ only if exactly one state variable is active at every layer.

Let $\text{LOS}(u, g)$ denote the rasterized Bresenham line cells strictly between u and g . The per-cell visibility/occlusion-depth surrogate is

$$C_{\text{occ}}(u) = |\text{LOS}(u, g) \cap O|. \quad (3)$$

A zero value indicates clear line of sight in this grid model. Let $B \subset V \setminus O$ be the set of non-obstacle cells within Chebyshev distance one of an obstacle. Cells in B are legal but risky.

HUBO Formulation

The total polynomial objective is

$$\begin{aligned} H(x) = & H_{\text{uniq}} + H_{\text{start}} + H_{\text{move}} + H_{\text{obs}} \\ & + H_{\text{occ}} + H_{\text{prox}} + H_{\text{goal}} + H_{\text{term}}. \end{aligned} \quad (4)$$

The structural terms are

$$H_{\text{uniq}} = \lambda_u \sum_{t=0}^{L-1} \left(\sum_{v \in V} x_{t,v} - 1 \right)^2, \quad (5)$$

$$H_{\text{start}} = \lambda_s (1 - x_{0,s}), \quad (6)$$

$$H_{\text{move}} = \lambda_m \sum_{t=0}^{L-2} \sum_{u \in V} \sum_{v \notin \mathcal{N}(u)} x_{t,u} x_{t+1,v}, \quad (7)$$

$$H_{\text{obs}} = \lambda_o \sum_{t=0}^{L-1} \sum_{v \in O} x_{t,v}. \quad (8)$$

The static visibility cost is linear:

$$H_{\text{occ}} = \lambda_{\text{occ}} \sum_{t=0}^{L-1} \sum_{u \in V} C_{\text{occ}}(u) x_{t,u}. \quad (9)$$

The sole cubic term is the sustained-proximity penalty:

$$\begin{aligned} H_{\text{prox}} = & \lambda_p \eta \sum_{t=1}^{L-2} \sum_{u \in B} \sum_{v \in B \cap \mathcal{N}(u)} \\ & \sum_{w \in B \cap \mathcal{N}(v)} x_{t-1,u} x_{t,v} x_{t+1,w}. \end{aligned} \quad (10)$$

Here $\eta = 3$. This term penalizes remaining in connected buffer cells for three consecutive layers. Thus the model is genuinely HUBO, but the higher-order degree comes from temporal buffer-risk modeling rather than from static visibility.

Quantity	Value
Grid size	8×8
Horizon L	20
Original binary variables	1,280
Total polynomial terms	118,344
Linear terms	1,280
Quadratic terms	112,672
Cubic terms	4,392
Obstacles	6
Buffer cells	24

Table 1: Primary six-obstacle HUBO instance.

The goal terms are

$$H_{\text{goal}} = \lambda_g \sum_{t=0}^{L-1} \sum_{v \in V} \frac{t+1}{L} \|p_v - p_g\|_{2x_{t,v}}, \quad (11)$$

$$H_{\text{term}} = \lambda_{\text{term}} \sum_{v \neq g} x_{L-1,v}. \quad (12)$$

The updated instance uses $\lambda_u = \lambda_s = \lambda_m = \lambda_o = 1000$, $\lambda_{\text{occ}} = 10$, $\lambda_p = 5$, $\lambda_g = 8$, $\lambda_{\text{term}} = 50$, and $\eta = 3$.

Feasibility-Aware Evaluation

A solver output is not scored only by total HUBO energy. Each binary sample is decoded and checked directly. We report: structural feasibility, meaning one-hot occupancy, correct start, legal transitions, and no obstacle occupancy; goal arrival, meaning $v_{L-1} = g$; and full-task success, meaning both conditions hold for the same decoded sample. Soft cost and implementation HUBO energy are then interpreted on valid decoded paths.

This distinction is essential for QUBO/BQM and QAOA-style workflows. A sample can have a low reduced-model energy while failing one-hot, motion, obstacle, or target-arrival checks. Such a sample is not a UAV plan. The evaluation protocol therefore reports feasibility rates and representative decoded trajectories, not energy alone.

Solvers and Implementation

All methods for all three instances were executed on the same Apple M2 Max platform with 32 GB unified memory. Timings cover solver calls and exclude plotting and file serialization; the QAOA-style QAOA time begins after candidate generation.

A* and RRT Baselines

A* searches directly on the grid with obstacle avoidance and a heuristic path cost. RRT-best is a randomized grid planner used as a sampling-style reference. These methods return explicit paths and are checked with the same feasibility evaluator. RRT-best uses 40 fixed-seed trials and best-cost selection; Table 2 reports the aggregate wall-clock time divided by 40.

Method	Feas.	Goal	Full	Len.	Turns	Buf.	Vis.	HUBO E	Time/run
A*	Yes	Yes	Yes	14	7	4	85.0%	-20742.26	0.028s
RRT	Yes	Yes	Yes	14	5	4	70.0%	-20700.19	0.0011s
QUAV-QAOA	Yes	Yes	Yes	14	1	3	80.0%	-20725.94	4.11s
CP-SAT exact	Yes	Yes	Yes	14	6	4	85.0%	-20752.43	1.02s
HUBO-SA	100.0%	16.0%	16.0%	14	6	4	85.0%	-20752.43	24.15s
QUBO/ <i>neal</i>	12.0%	8.0%	0.0%	18	12	2	20.0%	-20060.47	0.123s

Table 2: Unified same-platform comparison on the primary instance. Yes/no is a single output; percentages use 50 runs/reads. SA reports its best full-success path; *neal* reports its best-energy feasible non-goal sample. Time/run is mean wall-clock time; the *neal* value is amortized over one 50-read batch. Only *neal* samples the full penalized QUBO; the other methods construct or enforce some or all feasibility conditions before scoring.

QUAV-Style QAOA Baseline

QUAV maps UAV path segments to qubits and applies QAOA-style optimization with obstacle-buffer costs. Since no official public code was available in this workflow, we implemented a QUAV-style candidate-path selector following the published algorithmic description. It generates collision-free candidate paths, assigns a QUAV-style geometric cost using length, buffer steps, and turns, and uses a small QAOA-style one-hot selector over the candidates. This baseline is a same-platform reference only, not an exact reproduction of the authors’ implementation.

Trajectory-Level SA and *neal*

Trajectory-level SA searches over cell sequences and evaluates the implementation HUBO energy directly, including the cubic term. Its proposal operator preserves one-hot occupancy, start, legal motion, and obstacle avoidance by construction. In contrast, *neal* is a classical BQM simulated-annealing sampler (D-Wave Systems Inc. 2026b). Therefore the HUBO is first reduced to a QUBO/BQM using *dimod* (D-Wave Systems Inc. 2026a); cubic monomials require auxiliary-variable quadratization, following the standard reduction principle of Rosenberg (Rosenberg 1975), before sampling.

Exact CP-SAT Baseline

We implemented a CP-SAT model using OR-Tools (Perron and Furnon 2026). The model enforces exactly one cell per layer, the fixed start, final target arrival, obstacle avoidance, and legal transitions. Auxiliary binary variables linearize each cubic proximity monomial $y_{tuvw} = x_{t-1,u}x_{t,v}x_{t+1,w}$. CP-SAT requires integer coefficients, so the soft-objective coefficients are multiplied by 10^3 and rounded to the nearest integer. The model exactly minimizes this integer-scaled objective,

$$H_{\text{occ}} + H_{\text{prox}} + H_{\text{goal}} + H_{\text{term}} \quad (13)$$

over the hard-feasible set. In the solver implementation, the additive constant $\lambda_u L + \lambda_s = 21000$ is omitted. Consequently, the reported implementation energy is $\tilde{H} = H - 21000$. For a hard-feasible path with soft cost 247.57, this gives $\tilde{H} = -20752.43$. CP-SAT returned OPTIMAL; therefore, its solution provides a globally optimal reference for the implemented integer-scaled hard-feasible objective;

the table reports the corresponding path’s unrounded implementation HUBO energy.

Results

The quantitative results are organized around a primary instance and two obstacle-layout sensitivity instances. Table 1 summarizes the primary HUBO instance, including its grid, horizon, polynomial size, and obstacle geometry. Table 2 compares all six methods on that instance using feasibility, path quality, implementation HUBO energy, and mean runtime. Table 3 then reports the stochastic and exact results across all three fixed instances.

In Table 2, **Feas.** means one-hot occupancy, correct start, legal transitions, and obstacle avoidance; **Goal** means final-layer target arrival; and **Full** requires both. **Len.** counts non-hover moves, **Turns** direction changes excluding hovering, **Buf.** trajectory layers in obstacle-buffer cells, and **Vis.** layers with clear target line of sight. **HUBO E** is implementation energy with the constant 21,000 omitted.

The primary instance uses obstacles (3, 3), (3, 4), (4, 3), (4, 4), (2, 6), and (5, 1). The two sensitivity instances use respectively (2, 2), (6, 4) and (2, 2), (4, 3), (6, 4), with coordinates reported as (row, column); the three-obstacle case is nested by adding (4, 3) to the two-obstacle case. All three share the 8×8 grid, $L = 20$, start (0, 0), target (7, 7), objective weights, and solver budgets. The RRT trials use seeds 200–239; candidate generation uses A* seeds 0–79 and RRT seeds 1000–1079; the QAOA optimizer and *neal* use seed 42. For trajectory-SA run $r = 0, \dots, 49$, the search seed is $42 + r$; run 0 uses the greedy initialization, while runs $r = 1, \dots, 49$ use initialization seed $17r + 11$.

A*, RRT-best, and the QUAV-style QAOA selector return full-success paths on each sensitivity instance, and CP-SAT returns OPTIMAL on all three. SA remains structurally feasible by construction but its full-success rate changes from 16% on the primary instance to 84% and 66% on the two- and three-obstacle instances. The corresponding *neal* structural-feasibility rates are 12%, 26%, and 34%, while full-task success remains 0% in every case. These fixed-seed results are sensitivity checks, not distribution-level estimates. In particular, they do not isolate sampler behavior from the chosen terminal, structural, and quadratization penalties; the *neal* outcome characterizes the complete reduction-penalty-sampling pipeline used here.

Instance	Method	Feas.	Goal	Full	HUBO E
Primary (6 obs.)	SA	100.0% [92.9–100.0]	16.0% [8.3–28.5]	16.0% [8.3–28.5]	-20752.43
	<i>neal</i>	12.0% [5.6–23.8]	8.0% [3.2–18.8]	0.0% [0.0–7.1]	-20060.47
	CP-SAT (OPT.)	Yes	Yes	Yes	-20752.43
Sensitivity A (2 obs.)	SA	100.0% [92.9–100.0]	84.0% [71.5–91.7]	84.0% [71.5–91.7]	-20827.50
	<i>neal</i>	26.0% [15.9–39.6]	0.0% [0.0–7.1]	0.0% [0.0–7.1]	-20495.62
	CP-SAT (OPT.)	Yes	Yes	Yes	-20827.50
Sensitivity B (3 obs.)	SA	100.0% [92.9–100.0]	66.0% [52.2–77.6]	66.0% [52.2–77.6]	-20827.50
	<i>neal</i>	34.0% [22.4–47.8]	0.0% [0.0–7.1]	0.0% [0.0–7.1]	-20615.13
	CP-SAT (OPT.)	Yes	Yes	Yes	-20827.50

Table 3: Three-instance sensitivity results. Stochastic cells give the observed percentage and 95% Wilson interval over 50 fixed-seed runs or reads. SA reports its best full-success energy; *neal* reports its best-energy feasible sample; CP-SAT reports a single OPTIMAL output for the integer-scaled model. HUBO E omits the common constant 21,000.

Decoded Optimized Paths

Fig. 1 shows the actual decoded paths returned by each method on the same map. This figure is important because the numerical table alone does not reveal how the methods use the obstacle-buffer region or why different objectives prefer different routes. A*, RRT-best, QUAV-style QAOA, CP-SAT, and the best full-success SA trajectory all reach the target without collisions. The *neal* panel shows the best-energy structurally feasible decoded QUBO sample; it fails to reach the target, explaining why its full-task success rate is zero.

The CP-SAT baseline reports OPTIMAL for the integer-scaled model; its path has unrounded soft cost 247.57 and implementation HUBO energy -20752.43 . The best trajectory-SA path has the same reported energy, but SA reaches the target in only 8 of 50 runs. A* is fully successful, but its implementation HUBO energy is slightly worse because it optimizes a graph-search surrogate rather than the cubic proximity-aware objective used for final scoring.

The QUAV-style QAOA selector produces a valid shortest path with only one turn and the lowest QUAV-style geometric cost among its candidates. However, under the native HUBO objective its energy is -20725.94 , worse than the CP-SAT and best-SA paths. This illustrates a central point: a QUAV-style geometric objective and the native HUBO objective are not the same, even when both return collision-free paths. *neal* after quadratization remains the weakest method in this run: it has 6/50 structural feasibility and 4/50 goal arrival, but 0/50 full-task success.

Penalty Sweep

The penalty-weight sweep for *neal* shows that weak structural penalties permit low-energy but infeasible samples. In the updated run, structural feasibility was 0/15 for $\lambda \in \{50, 100, 300, 600\}$, 4/15 at $\lambda = 1000$, and 5/15 at $\lambda = 2000$. The best feasible soft cost improved from 915.4 to 590.6 between $\lambda = 1000$ and $\lambda = 2000$, still worse than A*, SA, and CP-SAT. This sweep varies the structural weights, not the terminal-goal weight or quadratization penalty, so it does not isolate goal-weight balance or sampler quality.

Discussion

The results show why the paper’s contribution is not a simple claim that quantum computing outperforms classical planning. On the primary instance, CP-SAT proves the optimum of the implemented integer-scaled hard-feasible objective, A* is fully successful, and trajectory-SA can match the CP-SAT path energy but with poor target-arrival reliability. The decoded paths in Fig. 1 make the solver differences explicit. The QUAV-style QAOA baseline is useful because it shows that a quantum-assisted geometric selector can produce a valid obstacle-free path on the same platform. However, the native HUBO objective distinguishes visibility, temporal buffer-risk, and terminal behavior in a different way. Therefore, same-platform decoded-path evaluation is necessary for fair comparison.

The exact CP-SAT formulation also clarifies the role of quantum-compatible solvers. If a QUBO sampler is used, it must overcome not only path planning but also one-hot, movement, obstacle, target, and quadratization constraints. The evaluation protocol exposes these failure modes by reporting structural feasibility, goal arrival, and full-task success separately.

The QUAV-style row in Table 2 should be interpreted carefully: it is a candidate-path selector over generated paths, not a native solution of the full $x_{t,v}$ HUBO. This candidate-path route is practical because feasibility is mostly handled before optimization, but it optimizes over a restricted candidate set. Thus the quantum-compatible components demonstrate formulation and evaluation pathways, not superiority over A* or CP-SAT. These classical methods remain the correctness anchors. If a quadratic sampler is used, the cubic term must first be reduced to quadratic form with auxiliary variables and additional penalties. This reduction is included as a conceptual conversion pathway and a tested failure mode rather than as a complete quadratization-accounting claim.

Limitations and Future Work

The model remains a grid abstraction without dynamics, control delays, or camera field-of-view constraints. A static target makes visibility linear; moving-target or occlusion-free tracking would require time-varying targets and hard

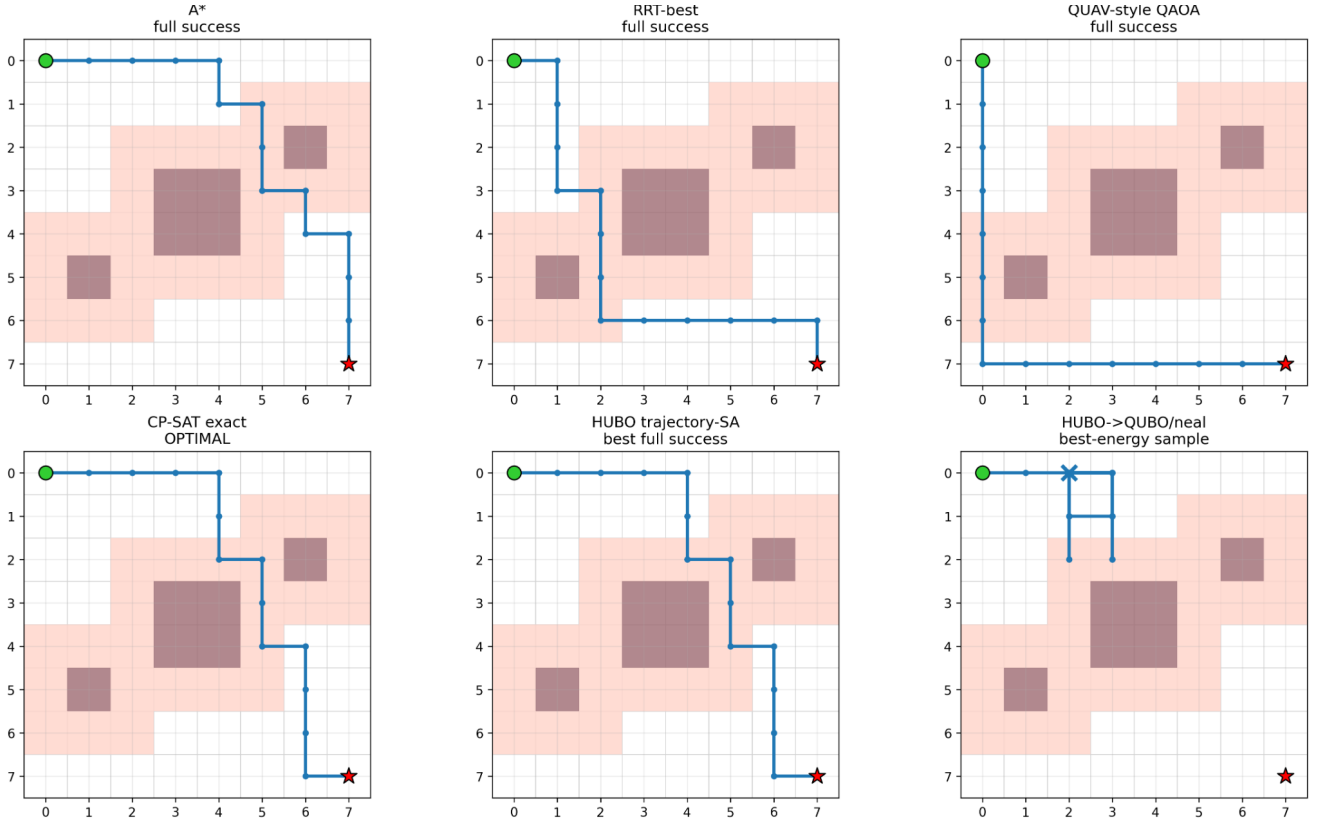


Figure 1: Decoded paths on the primary 8×8 , $L = 20$ evaluation instance. Dark cells are obstacles, light cells are obstacle-buffer cells, the green circle is the start, and the red star is the target. CP-SAT is globally optimal for the implemented integer-scaled hard-feasible objective. The structurally feasible `neal` sample does not reach the target, illustrating a reduction-penalty-sampling failure mode.

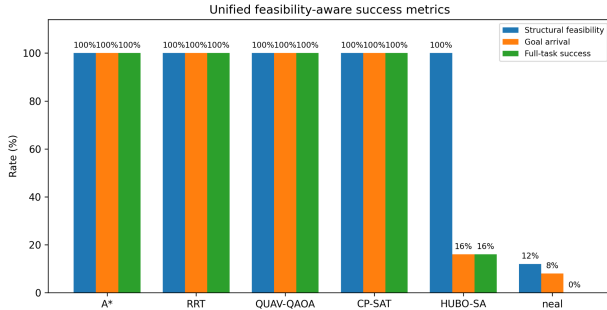


Figure 2: Feasibility-aware success metrics. CP-SAT certifies a full-success optimum for the integer-scaled model; `neal` has nonzero separate feasibility and goal-arrival rates but no full-task success.

line-of-sight constraints. With no public official QUAV implementation, our baseline evaluates the published concept rather than the full algorithm. Evidence covers three fixed maps that share grid size, horizon, start, target, and solver budgets; varying only obstacle count and placement is not a broad instance suite, so the rates are not distribution-level

estimates. Future work includes systematic variation of grid size, density, horizon, and endpoints, an ablation of the cubic proximity term against a quadratic buffer penalty, a matched QUAV implementation, moving targets, stronger candidate generation, and quadratization-penalty sensitivity.

Conclusion

This paper demonstrates a feasibility-aware HUBO/QUBO evaluation protocol for UAV obstacle-avoidance with visibility cost on one primary and two fixed sensitivity instances. A cubic term models sustained temporal proximity to obstacle-buffer cells. Comparing A*, RRT-best, QUAV-style QAOA, trajectory-SA, `neal`, and CP-SAT shows that geometric path quality, HUBO energy, and trajectory validity are different metrics. CP-SAT certifies an optimum for the integer-scaled model on all three instances, and the best observed SA path matches its reported energy in each, while quadratized `neal` has zero full-task success in all tested reads. The contribution is a certified fixed-instance protocol with limited obstacle-layout sensitivity, not quantum speedup or a general benchmark suite.

Code Availability

Code, figures, logs, and implementations are available at <https://github.com/alimohammadi44/uav-hubo-qiml-2026>.

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